

Layasaran Prabhu

layasaranp@gmail.com | 6379249239 | [linkedin/layasaran-p](https://www.linkedin.com/in/layasaran-p) | [github/LayasaranP](https://github.com/LayasaranP)

EDUCATION

B.Tech Mechanical Engineering and Minor in Computer Science Engineering
VELLORE INSTITUTE OF TECHNOLOGY

Tamil Nadu | Jul 2027

WORK EXPERIENCE

KONE | CLOUD ENGINEER INTERN

Chennai, Tamil Nadu | Jun 2025 – Jul 2025

- Designed and implemented a dashboard using **React** and **Tailwindcss** to visualize data retrieved from sensor and stored in **DynamoDB**, decreasing time to understand the health of elevator by over 10 times.
- Ensured only authorized employees have access to the application by creating an **AWS Lambda@Edge** function to intercept and sign valid requests.
- Devised and deployed the infrastructure in TypeScript through **AWS CDK**, then created efficient algorithms to process data from a custom **REST API**, so it could scale to handle millions of deliveries.

SKILLS

Languages: C, C++, Java, Python, SQL

AI & Machine Learning: PyTorch, Keras, Hugging Face

LLM Orchestration: LangChain, LangGraph, DeepAgents, Vector Databases (Pinecone, Qdrant, ChromaDB)

Web Development: React, Next.js, Tailwind CSS, Express.js, FastAPI, Spring Boot (Basics)

Tools: Git, AWS, Docker, Vercel, Jira.

PROJECTS

Bearing Fault Diagnosis | PYTORCH, PIL, LIBROSA, SIGNAL PROCESSING, TRANSFER LEARNING

- Engineered a signal-to-image conversion pipeline that transforms raw 1D vibration signals into 150×150 RGB images by extracting & encoding time-domain features (15: RMS, kurtosis, crest factor, etc.), frequency-domain spectrogram characteristics (65: spectral bands, entropy, harmonics), and operating conditions via color-gradient mapping — enabling powerful 2D CNN application for bearing fault detection
- Built & benchmarked a PyTorch transfer learning system comparing EfficientNet-B4, DenseNet-121, and ResNet50; achieved **97.6%** accuracy across CWRU, Paderborn, and custom experimental datasets — enhanced with Grad-CAM heatmaps for interpretability and combined signal processing + computer vision for robust industrial predictive maintenance

Hybrid Log Message Classification | PYTHON, FASTAPI, SENTENCE TRANSFORMERS, SCIKIT-LEARN

- Designed a hybrid log classification pipeline combining fast regex rules for structured patterns, Sentence Transformers + Logistic Regression for accurate supervised classification, and LLM fallback for ambiguous/rare cases — deployed as a FastAPI REST service (with Swagger UI) that processes CSV logs and returns enriched predictions with `target_label`
- Balanced speed (rules), accuracy (ML), and robustness (LLM) while extending regex coverage to 25+ patterns, comparing multiple LLMs, adding detailed F1-score / confusion matrix evaluation, and publishing an interactive demo — directly applicable to observability, SRE, and anomaly detection systems

COURSEWORKS

Ultimate RAG Bootcamp (LangChain, LangGraph)

2025

Key Areas: Traditional RAG, Agentic RAG, Multi-agent Systems, LangGraph State Management, LangSmith Observability.